

Exact Certificates for Unimodality of Forest Independence Polynomials

A consolidated proof manuscript and reproducible supplement

Tong Zhang

ZhangZenoZhang@gmail.com

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Abstract

We assemble two certificate-assisted arguments for the independence sequences of finite simple forests. The finite-order argument covers every forest on at most 60 vertices by structural counting inequalities, 670 rational separating inequalities, and 1907 nonnegative rational linear certificates. A separate hard-core argument, supplied with a frozen dependency chain, asserts unimodality for all forests on at least 59 vertices. Their scopes overlap, so that no order is omitted. We give the finite-order structural proof, the conditional-mixture reduction and new matching-count constraints for the large-order argument, and a precise map to the full inherited mathematical and computational supplements. Acceptance uses integer or directed rational arithmetic; numerical optimizers are not part of the verification. This is a mathematical proof submission for review, not a claim of completed Lean formalization or prior journal acceptance. A rational obstruction at order 100 concerns only the finite-order relaxation and is not a graph counterexample.

Keywords. Forest; independent set; unimodality; hard-core model; exact certificate; computer-assisted proof.

1 The statement, scope, and proof architecture

For a finite simple graph F , write

$$I_F(x) = \sum_{r=0}^{\alpha(F)} c_r(F)x^r, \quad c_r(F) = |\{S \subseteq V(F) : S \text{ independent, } |S| = r\}|.$$

A finite sequence is unimodal when it weakly increases to some index and weakly decreases thereafter. We include the empty forest, with polynomial 1, as well as isolated vertices and disconnected forests. Throughout, n denotes the number of vertices; it never denotes the matching number or a coefficient index.

The tree and forest questions originate in Alavi, Malde, Schwenk, and Erdős [1] and appear as Erdős Problem 993 and Justin Sun Prize Problem JSP-000826 [2, 3]. The forest assertion cannot be inferred merely by multiplying arbitrary unimodal tree polynomials: unimodality is not generally preserved by convolution. Our two inputs each quantify over forests directly.

Theorem 1.1 (Consolidated assertion submitted for review). *The independence sequence of every finite simple forest is unimodal.*

The proof is organized through the following two propositions.

Proposition 1.2 (Finite-order part). *Every finite simple forest on at most 60 vertices has a unimodal independence sequence.*

Proposition 1.3 (Large-order part). *Every finite simple forest on at least 59 vertices has a unimodal independence sequence.*

Proof of Theorem 1.1 from the two propositions. If $n \leq 60$, use Proposition 1.2. If $n \geq 61$, use Proposition 1.3. The two propositions also independently cover $n = 59, 60$. No induction across these orders and no closure under disjoint union is needed. $\square$

Proposition 1.2 is developed below with its exact certificate interface. Proposition 1.3 uses the large-order manuscript and frozen dependency chain distributed as Supplement A. We reproduce its final extension and the principal analytic interface in Sections 3–6. The full analytic proofs and earlier thresholds are integral parts of the supplement, not assumptions supplied by a numerical success flag. The historical wording “conditional on inherited structural lemmas” in the input releases refers to those stated dependencies. To accept the consolidated theorem, a reviewer must accept their proofs as well as the finite arithmetic; two overlapping numerical ranges alone are not a proof.

Fang, Lu, Nevo, Yao, and Zheng [4] prove unimodality for every sufficiently large forest. Their Theorem 1.1 states the existence of an absolute threshold; it does not supply the value 59 used here. We do not substitute that existence statement for the explicit threshold chain, nor claim priority over their asymptotic result. Counterexamples to log-concavity of tree independence sequences [6, 7] do not contradict the unimodality assertion. We make no assertion of log-concavity of the full sequence.

Reading the supplement. Supplement A is the original `forest_threshold_59_audited_handoff.zip`; its nested archives preserve the stages 80, 99, 130, 170, 350, 500, 900. Supplement B is `forest_n60_extension_and_n100_gap.zip`. A separate Chinese finite-order proof supplied with the files is byte-identical to its copy in Supplement B. The distribution includes those original bytes, a flattened map of mathematical source documents, the present English manuscript, and fresh audit evidence. Supplementary Chinese texts retain their original notation and historical scope statements.

2 A finite certificate theorem for all forests of order at most sixty

For a finite simple graph F , let $c_r(F)$ denote the number of independent sets of size r , and let

$$I_F(z) = \sum_{r=0}^{\alpha(F)} c_r(F) z^r$$

be its independence polynomial. We use the non-strict convention for unimodality: a sequence is unimodal if it is nondecreasing up to some index and nonincreasing thereafter. Thus a plateau at the maximum is allowed. The result in this section concerns arbitrary forests, including disconnected forests and forests with isolated vertices.

Theorem 2.1 (Finite-order forest unimodality). *If F is a forest with at most 60 vertices, then the coefficient sequence of $I_F(z)$ is unimodal.*

The proof has two parts. Structural inequalities place the independent-set counts of every forest in an explicitly defined real polyhedron. A finite collection of rational certificates then establishes the required signs of successive coefficient differences throughout the relevant

parameter domain. The certificate stage does not enumerate forests. It is, however, an essential part of the proof: the result is not asserted to have a certificate-free closed-form proof. Neither Theorem 2.1 nor the argument below asserts log-concavity of the entire coefficient sequence, unimodality for all orders, or a corresponding theorem up to order 100.

Throughout this section, binomial coefficients are defined by

$$\binom{b}{r} = 0 \quad \text{unless } b, r \text{ are integers with } 0 \leq r \leq b.$$

For polynomials with real coefficients, $P \leq_{\text{coeff}} Q$ means that every coefficient of $Q - P$ is nonnegative. We write $x_+ = \max(x, 0)$ and use $\mathbf{1}_A$ for the indicator of a condition A .

2.1 A decomposition relative to a maximum independent set

Let F be a forest on n vertices. Write $a = \alpha(F)$ and let v be its matching number. Leaf deletion gives

$$a + v = n, \quad \delta := a - v \geq 0. \quad (1)$$

For completeness, if u is a leaf with neighbour w , a maximum independent set may be chosen to contain u : replace w by u if necessary. Consequently $\alpha(F) = 1 + \alpha(F - \{u, w\})$. A maximum matching may likewise be chosen to contain uw , by replacing an edge incident with w by uw when necessary. Its size is therefore $1 + \nu(F - \{u, w\})$. An isolated vertex increases the independence number by one and leaves the matching number unchanged. These observations prove the first identity in (1) by induction. The inequality follows from $2v \leq n$.

Fix a maximum independent set I and put $Y = V(F) \setminus I$, so that $|I| = a$ and $|Y| = v$. For any fixed maximum matching, an independent set contains at most one endpoint of each matched edge, and there are $\delta = n - 2v$ unmatched vertices. Since $|I| = v + \delta$, equality holds in all of these restrictions. Hence I contains every unmatched vertex and exactly one endpoint of every matched edge. In particular, the matching saturates Y and matches it into I .

For an independent set $J \subseteq Y$, define

$$j = |J|, \quad m(J) = |I \setminus N_I(J)|.$$

Let $t_{j,m}$ count the independent subsets J of Y for which these parameters equal j, m . The set $J \cup (I \setminus N_I(J))$ is independent, so $j + m \leq a$. By convention $t_{0,a} = 1$, and all other $t_{0,m}$ vanish. Every independent set of F is uniquely obtained by first choosing its intersection J with Y and then choosing an arbitrary subset of $I \setminus N_I(J)$. Thus

$$I_F(z) = (1 + z)^a + \sum_{j=1}^v \sum_{m=0}^{a-j} t_{j,m} z^j (1 + z)^m. \quad (2)$$

In particular,

$$c_r = \binom{a}{r} + \sum_{j=1}^v \sum_{m=0}^{a-j} \binom{m}{r-j} t_{j,m}. \quad (3)$$

No maximality requirement is placed on J ; all independent sets are included in this identity.

Define

$$T_j = \sum_m t_{j,m}, \quad M_j = \sum_m m t_{j,m}, \quad W_j = \binom{v}{j}. \quad (4)$$

The basic constraints are

$$t_{j,m} \geq 0, \quad T_1 = v, \quad T_j \leq W_j \quad (2 \leq j \leq v). \quad (5)$$

All certificate arguments below remain valid when the $t_{j,m}$ are arbitrary nonnegative real numbers satisfying the displayed constraints. Integrality is used in deriving some structural inequalities for actual forests, but is not imposed in the subsequent linear relaxation.

2.2 Coefficient bounds and the first cross-layer inequalities

Lemma 2.2 (Path lower bound and matching upper bound). *For every forest of order n and independence number a , with $v = n - a$,*

$$\binom{n-r+1}{r} \leq c_r \quad (r \geq 0), \quad (6)$$

and

$$I_F(z) \leq_{\text{coeff}} (1+z)^{a-v}(1+2z)^v. \quad (7)$$

Proof. The number of independent r -sets of the path on n vertices is $p_{n,r} = \binom{n-r+1}{r}$. A leaf u with neighbour w gives

$$I_F = I_{F-u} + zI_{F-\{u,w\}},$$

and $p_{n,r} = p_{n-1,r} + p_{n-2,r-1}$. Induction therefore proves (6) when a leaf is present. If u is isolated, $I_F = (1+z)I_{F-u}$; the induction bound and $p_{n-1,r-1} \geq p_{n-2,r-1}$ give the same conclusion. The empty and one-vertex forests start the induction. Coefficients beyond the degree are handled by the binomial convention.

For the upper bound, $m \leq a - j$ in (2), and the layer mass is at most $\binom{v}{j}$. Consequently

$$\begin{aligned} I_F(z) &\leq_{\text{coeff}} \sum_{j=0}^v \binom{v}{j} z^j (1+z)^{a-j} \\ &= (1+z)^{a-v} (1+2z)^v. \end{aligned}$$

□

Lemma 2.3 (Private-neighbour inequality). *For $2 \leq r \leq v$ and every integer $h \geq 0$,*

$$\sum_m ((r-1)m + a - r + 1 - rh)_+ t_{r,m} \leq (v-r+1) \sum_m (m-h)_+ t_{r-1,m}. \quad (8)$$

Proof. Fix an independent r -set $J \subseteq Y$, put $m = m(J)$ and $d = a - m$, and consider the bipartite forest induced by $J \cup N_I(J)$. Each vertex of J has a neighbour in I , since otherwise I would not be maximum. If this forest has $b \geq 1$ components and p vertices of degree one on its I side, its number of edges is $r + d - b$, while summing degrees on that side gives at least $p + 2(d-p)$. Therefore

$$p \geq d - r + b \geq d - r + 1.$$

For $u \in J$, let p_u be the number of its private neighbours in I , that is, neighbours of u adjacent to no other member of J . Then

$$m(J \setminus \{u\}) = m + p_u, \quad \sum_{u \in J} p_u \geq d - r + 1. \quad (9)$$

It follows that

$$\sum_{u \in J} (m + p_u - h)_+ \geq \left(r(m-h) + \sum_{u \in J} p_u \right)_+ \geq ((r-1)m + a - r + 1 - rh)_+.$$

Sum over J . Each independent $(r-1)$ -set of Y is obtained by deleting one vertex from at most $v - r + 1$ independent r -sets. The resulting reverse count proves (8). □

Lemma 2.4 (Matching-based extension inequality). *Put*

$$H_{r,l} = \sum_m \binom{m}{l} t_{r,m}, \quad H_{0,l} = \binom{a}{l}.$$

Whenever the factor on the right is nonnegative,

$$(r+1)H_{r+1,l} \leq (v-r-\max(0, l-\delta))H_{r,l}. \quad (10)$$

Proof. Fix a set $L \subseteq I$ of size l . At most δ vertices of I are unmatched by a fixed maximum matching. The matched vertices of L therefore exclude at least $\max(0, l-\delta)$ distinct vertices of Y . Among the remaining vertices, an independent r -set has at most $v-r-\max(0, l-\delta)$ extensions. Double-count the pairs consisting of such an r -set and one extending vertex, and then sum over L . An independent set J contributes $\binom{m(J)}{l}$ choices of L , giving precisely (10). $\square$

2.3 Choosing a maximum independent set with a sparse complement

The preceding formulas hold for every maximum independent set. The remaining inequalities are used with a single maximum independent set chosen as follows.

Lemma 2.5 (Sparse-complement selection). *A forest has a maximum independent set I whose complement Y satisfies*

$$e(F[Y]) \leq E := \max(0, \delta - 1). \quad (11)$$

If $v = |Y| \geq 1$, then also $e(F[Y]) \leq v - 1$.

Proof. Fix a maximum matching and a bipartite colouring of each component of F . Contract each matched edge to one paired node, and retain each unmatched vertex as a terminal. The contracted graph is a forest. Every maximum independent set contains all terminals and selects exactly one endpoint from each matched pair. Label a paired node by the original bipartite colour of its selected endpoint, and label a terminal by its own colour.

Consider an edge of the contracted forest. Its corresponding original edge joins opposite bipartite colours. If the two node labels agree, the original edge has exactly one endpoint in I . If they disagree, it has either both endpoints in I or both in Y ; the former is excluded by independence. Thus edges between different labels correspond precisely to edges of $F[Y]$. No such edge is incident with a terminal.

Choose a maximum independent set minimizing $e(F[Y])$. Suppose a monochromatic connected block in the contracted forest has no terminal and has a boundary edge. Swap the selected and unselected endpoints in every matched pair belonging to this block. Internal edges still have exactly one selected endpoint. Every boundary edge, previously an edge with both endpoints in Y , now has exactly one selected endpoint. Matched edges still contribute one selected endpoint each. Hence the new set is independent, has the same size, and has strictly fewer edges in its complement, a contradiction.

In an original component containing $\delta_c > 0$ terminals, every monochromatic block consequently contains a terminal. There are at most δ_c such blocks, and contracting them gives a tree, so there are at most $\delta_c - 1$ edges between blocks. In a component without terminals there can be only one monochromatic block, hence no such edges. Summing over components proves (11). The final assertion follows because $F[Y]$ is itself a forest. $\square$

From now on, fix the set supplied by Lemma 2.5. Define

$$D_0 = \min(E, v - 1), \quad e = e(F[Y]) = \binom{v}{2} - T_2. \quad (12)$$

The last identity is exact: a two-element subset of Y fails to be independent precisely when it is an edge. All constraints below refer to this same I , this same Y , and this same value of e .

2.4 Two-sided information from deletion of a vertex

For an independent r -set $J \subseteq Y$, retain the notation $m = m(J)$, $d = a - m$, and put $L = d - r + 1$. Since $r + m \leq a$, we have $L \geq 1$. Moreover, applying maximality of the independence number to $J \setminus \{u\}$ gives $(r - 1) + (m + p_u) \leq a$. Together with (9) and the fact that private neighbours are distinct, this yields

$$0 \leq p_u \leq L, \quad L \leq \sum_{u \in J} p_u \leq d. \quad (13)$$

For $2 \leq r \leq v$ and an integer $h \geq 0$, define

$$b_{r,h}(m) = \begin{cases} r, & m \geq h, \\ 1, & m < h \text{ and } rh \leq a + (r - 1)m, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Lemma 2.6 (Lower deletion threshold). *For every $2 \leq r \leq v$ and integer $h \geq 0$,*

$$\sum_m b_{r,h}(m) t_{r,m} \leq (v - r + 1) \sum_{m \geq h} t_{r-1,m}. \quad (15)$$

Proof. If $m \geq h$, all r deletions from J give an available-vertex count at least h . Otherwise let $t = h - m \geq 1$. The condition in the second line of (14) is $rt \leq d$. If every deletion had $p_u < t$, integrality would imply

$$\sum p_u \leq r(t - 1) \leq d - r < L,$$

contrary to (13). Thus at least one deletion has the required count. Summing this lower bound over J and using at most $v - r + 1$ extensions for each parent set proves the inequality. $\square$

For an independent $(r - 1)$ -set $A \subseteq Y$, the number of unselected vertices forbidden by its edges is at most the total number e of edges of $F[Y]$. Hence A has at least

$$s_r = \max(0, v - r + 1 - E) \quad (16)$$

extensions to an independent r -set.

To bound deletion sums in the other direction, write

$$d = qL + \rho, \quad 0 \leq \rho < L.$$

The identity $rL - d = (r - 1)(d - r) \geq 0$ gives $q \leq r$; if $q = r$, then $\rho = 0$. Define

$$U_{r,h}(m) = q(m + L - h)_+ + \begin{cases} (m + \rho - h)_+ + (r - q - 1)(m - h)_+, & q < r, \\ 0, & q = r, \end{cases} \quad (17)$$

and

$$V_{r,h}(m) = \begin{cases} r, & h \leq m, \\ \min(r, \lfloor d/(h - m) \rfloor), & m < h \leq m + L, \\ 0, & h > m + L. \end{cases} \quad (18)$$

Lemma 2.7 (Upper deletion bounds). *For $2 \leq r \leq v$ and integer $h \geq 0$,*

$$s_r \sum_m (m - h)_+ t_{r-1,m} \leq \sum_m U_{r,h}(m) t_{r,m}, \quad (19)$$

$$s_r \sum_{m \geq h} t_{r-1,m} \leq \sum_m V_{r,h}(m) t_{r,m}. \quad (20)$$

Proof. For fixed J , consider the feasible release vector $(p_u)_{u \in J}$ in (13). The function $p \mapsto (m + p - h)_+$ is convex and nondecreasing. Under the upper constraints $0 \leq p_u \leq L$ and $\sum p_u \leq d$, its sum is maximized by concentrating mass: q coordinates equal L , at most one further coordinate equals ρ , and the others are zero. This follows, for example, by repeatedly transferring mass between two coordinates strictly inside $[0, L]$ toward an endpoint, choosing a direction that does not decrease the convex sum. Monotonicity permits total mass d . The resulting maximum is (17).

For the indicator sum, all deletions qualify when $h \leq m$. If $0 < h - m \leq L$, each qualifying coordinate consumes at least $h - m$ of the total mass at most d , giving the middle bound in (18). If $h - m > L$, none qualifies.

Now sum either deletion statistic over all independent r -sets. The upper bounds just obtained give the right sides of (19) and (20). Counting the same incidence relation from its $(r - 1)$ -set side gives the left sides, because every such set has at least s_r extensions. When $s_r = 0$, both inequalities remain valid and are automatic consequences of nonnegativity. $\square$

2.5 Available-vertex means and the complement's actual edge count

Lemma 2.8 (Coverage union bound). *For $2 \leq r \leq v$,*

$$\sum_m (a - m) t_{r,m} \leq \binom{v-1}{r-1} \sum_m (a - m) t_{1,m}. \quad (21)$$

Proof. For each independent r -set $J \subseteq Y$, its covered vertices in I form a union of the neighbourhoods of its members. Its cardinality is at most the sum of those neighbourhood sizes. A fixed vertex of Y belongs to at most $\binom{v-1}{r-1}$ such sets. Summing the union bound therefore gives the stated inequality. $\square$

For $2 \leq r \leq v$, put

$$\beta_r = \binom{v-2}{r-1} - (a-2) \binom{v-2}{r-2}, \quad C_r^0 = a \binom{v-2}{r} + (\delta+1) \binom{v-2}{r-1}. \quad (22)$$

Lemma 2.9 (Mean bound retaining the actual edge count). *With $e = e(F[Y])$ as in (12),*

$$M_r \geq C_r^0 + \beta_r e \quad (2 \leq r \leq v). \quad (23)$$

Proof. For $i \in I$, let $d_i = |N(i) \cap Y|$. For integer $0 \leq d \leq v$, the function $d \mapsto \binom{v-d}{r}$ is discretely convex: its forward differences are $-\binom{v-d-1}{r-1}$ and are nondecreasing. The line through its values at $d = 1, 2$ is therefore a lower support line on the integer domain. Explicitly,

$$\binom{v-d_i}{r} \geq \binom{v-2}{r} + (2-d_i) \binom{v-2}{r-1}.$$

The number of edges between I and Y is at most $n - 1 - e$. Consequently

$$\begin{aligned} \sum_{i \in I} \binom{v-d_i}{r} &\geq a \binom{v-2}{r} + (2a - n + 1 + e) \binom{v-2}{r-1} \\ &= C_r^0 + e \binom{v-2}{r-1}. \end{aligned}$$

The sum on the left counts pairs consisting of any r -subset of Y and an available vertex of I , without yet requiring the subset to be independent.

For an edge uw of $F[Y]$, both u and w have nonempty neighbourhoods in I , and those neighbourhoods are disjoint, since a common neighbour would produce a triangle. Every r -set containing u, w therefore has at most $a - 2$ available vertices of I . There are $\binom{v-2}{r-2}$ candidate sets containing this edge. A union bound over the e edges shows that deleting all nonindependent subsets from the preceding count loses at most $e(a - 2)\binom{v-2}{r-2}$. Subtracting this amount proves the lemma. $\square$

The coefficient β_r may be negative. Accordingly we do not replace e independently by an endpoint of its allowed interval for each r . We retain the common variable and then substitute the exact identity $e = \binom{v}{2} - T_2$; this is the source of the linear row used in the certificates.

2.6 Coefficient bounds for a forest with a fixed edge count

Lemma 2.10 (Edge-count coefficient bounds). *For a forest G on $v \geq 1$ vertices and with e edges,*

$$i_r(G) \geq \binom{v}{r} - e \binom{v-2}{r-2}, \quad (24)$$

where $i_r(G)$ counts its independent r -sets, and

$$I_G(z) \leq_{\text{coeff}} (1+z)^{v-1} + z(1+z)^{v-e-1}. \quad (25)$$

Proof. The first bound deletes from the $\binom{v}{r}$ subsets all those containing an edge. Each edge occurs in $\binom{v-2}{r-2}$ subsets, so the union bound proves the claim, including cases when its right side is negative.

For the upper bound, write $B(v, e) = (1+z)^{v-1} + z(1+z)^{v-e-1}$. If $e = 0$, equality holds; orders at most two are immediate. For $v \geq 3$ and $e > 0$, choose a leaf u with neighbour w of degree d . The two smaller forests have parameters $(v-1, e-1)$ and $(v-2, e-d)$. Induction and leaf deletion give

$$I_G(z) \leq_{\text{coeff}} B(v-1, e-1) + zB(v-2, e-d).$$

A direct subtraction gives

$$B(v, e) - B(v-1, e-1) - zB(v-2, e-d) = z^2((1+z)^{v-3} - (1+z)^{v-e+d-3}).$$

The remaining forest on $v-2 \geq 1$ vertices has $e-d \leq v-3$ edges, so the second exponent is nonnegative. Since $e \geq d$, it does not exceed the first exponent. The displayed difference has nonnegative coefficients, completing the induction. $\square$

Apply the lemma to $G = F[Y]$, whose independent-set coefficients are T_r . From (25),

$$T_r \leq \binom{v-1}{r} + \binom{v-e-1}{r-1}. \quad (26)$$

For $2 \leq r \leq v$, the function on integer $0 \leq e \leq v-1$ given by

$$g_r(e) = \binom{v-1}{r-1} - \binom{v-e-1}{r-1}$$

is concave, satisfies $g_r(0) = 0$, and has successive increments $\binom{v-e-2}{r-2}$, which are nonincreasing. If $D_0 \geq 1$, define

$$G_r = \binom{v-1}{r-1} - \binom{v-D_0-1}{r-1}. \quad (27)$$

The chord bound gives $g_r(e) \geq eG_r/D_0$ for $0 \leq e \leq D_0$. Substituting (12) in the lower bound and in this linearized upper bound yields

$$-T_r + \binom{v-2}{r-2} T_2 \leq -\binom{v}{r} + \binom{v-2}{r-2} \binom{v}{2}, \quad (28)$$

$$D_0 T_r - G_r T_2 \leq D_0 \binom{v}{r} - G_r \binom{v}{2}. \quad (29)$$

When $D_0 = 0$, the implemented replacement for (29) is the ordinary bound $T_r \leq \binom{v}{r}$. This replacement is weaker than all information available in that case, but remains valid and is sufficient for the finite certificates.

2.7 Reduction to successive differences on a finite parameter domain

Set

$$\Delta_k = c_k - c_{k-1}, \quad B_{m,j}(k) = \binom{m}{k-j} - \binom{m}{k-j-1}. \quad (30)$$

The exact coefficient identity gives

$$\Delta_k = B_{a,0}(k) + \sum_{j,m} B_{m,j}(k) t_{j,m}. \quad (31)$$

Lemma 2.11 (Universal decreasing tail). *For a forest of independence number a ,*

$$(k+1)c_{k+1} \leq 2(a-k)c_k. \quad (32)$$

In particular, with

$$K(a) = \left\lfloor \frac{2a+1}{3} \right\rfloor, \quad (33)$$

the sequence $c_{K(a)}, c_{K(a)+1}, \dots, c_a$ is nonincreasing.

Proof. Given an independent k -set S , let $A(S)$ be the vertices that can be added to it. The induced forest on $A(S)$ has independence number at most $a-k$, since any of its independent sets can be joined to S . A bipartite graph has an independent set containing at least half its vertices, so $|A(S)| \leq 2(a-k)$. Count pairs (S, u) with $u \in A(S)$. Every independent $(k+1)$ -set occurs $k+1$ times, proving (32). The factor $2(a-k)/(k+1)$ is at most one when $k \geq \lceil (2a-1)/3 \rceil = K(a)$. $\square$

For each $1 \leq k < K(a)$ it is sufficient to prove one of the statements

$$\Delta_k > 0, \quad \Delta_{k+1} \leq 0, \quad \Delta_k \leq 0 \implies \Delta_{k+1} \leq 0. \quad (34)$$

Indeed, once a nonpositive difference occurs, these alternatives prevent a later positive difference before the tail. The first alternative cannot apply at an index where the difference is already nonpositive, while the other two give the required next sign. Lemma 2.11 supplies all remaining signs. Also $c_0 = 1$ and $c_1 = n$.

The empty forest and forests with no edges have independence polynomial 1 or $(1+z)^n$, respectively, and are handled directly. For forests with at least one edge and order at most 60, all required triples lie in the explicitly regenerated domain

$$\mathcal{P}_{60} = \left\{ (n, a, k) : \begin{array}{l} 2 \leq n \leq 60, \quad \lceil n/2 \rceil \leq a \leq n-1, \\ 1 \leq k < \lfloor (2a+1)/3 \rfloor \end{array} \right\}. \quad (35)$$

It contains 17 100 triples. Empty ranges for small values of a require no certificate. These triples are scalar parameter choices, not representatives of graph isomorphism classes.

2.8 Exact certificate identities and the complete finite check

Direct criteria. Let

$$q_{a,v}(r) = \sum_{j=0}^{\min(v,r)} \binom{v}{j} 2^j \binom{a-v}{r-j}, \quad (36)$$

the coefficient of z^r in the upper-bound polynomial of Lemma 2.2. The comparisons

$$\binom{n-k+1}{k} > q_{a,v}(k-1) \quad \text{or} \quad q_{a,v}(k+1) \leq \binom{n-k+1}{k}$$

prove, respectively, $\Delta_k > 0$ or $\Delta_{k+1} \leq 0$.

The basic layer masses give further direct bounds. For each j define

$$L_j(r) = \begin{cases} \min_{0 \leq m \leq a-1} B_{m,1}(r), & j = 1, \\ \min(0, \min_{0 \leq m \leq a-j} B_{m,j}(r)), & j \geq 2, \end{cases}$$

$$U_j(r) = \begin{cases} \max_{0 \leq m \leq a-1} B_{m,1}(r), & j = 1, \\ \max(0, \max_{0 \leq m \leq a-j} B_{m,j}(r)), & j \geq 2. \end{cases}$$

Then

$$B_{a,0}(r) + \sum_{j=1}^v W_j L_j(r) \leq \Delta_r \leq B_{a,0}(r) + \sum_{j=1}^v W_j U_j(r). \quad (37)$$

The absence of the extra zero in the first layer is justified by $T_1 = v$ exactly. A strictly positive lower bound at $r = k$ or a nonpositive upper bound at $r = k + 1$ proves the corresponding alternative in (34).

For a rational number $s \geq 0$, the same upper-bound rule can be applied to $B_{m,j}(k+1) - sB_{m,j}(k)$. A nonpositive resulting bound proves

$$\Delta_{k+1} \leq s\Delta_k, \quad (38)$$

which proves the implication in (34). These are the one-parameter rational comparison certificates called **separator** in the data.

General linear certificates. Normalize the variables by $x_{j,m} = t_{j,m}/W_j$. Their layer masses are at most one, and the first layer mass is exactly one. For every valid integer inequality

$$\sum_{j,m} a_{j,m}^\rho t_{j,m} \leq b^\rho,$$

define its scale by

$$N_\rho = \max \left(1, |b^\rho|, \max_{j,m} |W_j a_{j,m}^\rho| \right), \quad (39)$$

and write the corresponding normalized row as $A^\rho x \leq b_\rho$, where $A_{j,m}^\rho = W_j a_{j,m}^\rho / N_\rho$ and $b_\rho = b^\rho / N_\rho$. Normalize the target affine function in the same way, obtaining

$$Q(x) = q_0 + \sum_{j,m} q_{j,m} x_{j,m}.$$

If $\lambda_\rho \geq 0$ and $\eta \in \mathbb{R}$, set

$$R_{j,m} = q_{j,m} - \sum_\rho \lambda_\rho A_{j,m}^\rho - \eta \mathbf{1}_{j=1}.$$

The following elementary identity is the entire certificate principle:

$$Q(x) \leq q_0 + \sum_{\rho} \lambda_{\rho} b_{\rho} + \eta + \sum_{j=1}^v \max(0, \max_m R_{j,m}). \quad (40)$$

To see this, subtract the nonnegative combination of valid rows and use the first-layer equality for the η term. On each remaining layer, nonnegativity and mass at most one bound its residual contribution by the nonnegative maximum of its residual coefficients. No assertion about linear-programming optimality is needed.

There are three linear certificate types. A **positive** certificate uses the target $-\Delta_k$ and makes the right side of (40) strictly negative. A **negative** certificate uses Δ_{k+1} and makes the bound nonpositive. A **conditional** certificate has the latter target and is permitted to add the row $\Delta_k \leq 0$. It therefore proves exactly the implication in (34). That assumption cannot be discarded while retaining an unconditional negative target claim.

Exact arithmetic and exhaustive parameter coverage. Each stored linear certificate gives a positive integer denominator D_c , integer row-weight numerators, and an integer numerator for the unrestricted equality weight. The principal checker `verify.py` independently constructs the unweighted integer rows, computes a common multiple of their normalization denominators, and checks (40) using integers. It regenerates every triple in (35); it rejects missing, duplicate, or extraneous triples, invalid row parameters, negative inequality multipliers, and conditional assumptions with the wrong index. Direct and one-parameter comparisons are likewise performed exactly.

A second arithmetic implementation, `verify_grid.py`, uses the weighted rows from `kernel.py`. It rounds target coefficients and right sides upward and row coefficients downward on an integer grid. Nonnegative variables and nonnegative row multipliers make this an outward, conservative bound. This second implementation checks all 1 907 linear certificates; complete parameter coverage and the 670 one-parameter comparisons are checked by the principal verifier. Both implementations use the Python standard library, without a graph catalogue, an optimizer, or floating-point feasibility tolerances.

The complete partition obtained by rerunning these checks is shown in Table 1. The matching-upper/path-lower test for a nonpositive next difference is available in the verifier, but accounts for no additional cases in this partition.

Certificate or direct criterion	Number of triples
Path lower bound exceeds the preceding matching upper bound	6 724
Direct positive lower bound for Δ_k	5 709
Direct nonpositive upper bound for Δ_{k+1}	2 090
One-parameter rational comparison	670
positive linear certificate	603
negative linear certificate	158
conditional linear certificate	1 146
Total	17 100

Table 1: Complete parameter coverage for Theorem 2.1.

The data contain 2 577 stored triples: 670 comparison parameters and 1 907 linear certificates. The latter contain 142 017 nonzero row-weight terms. There are at most 610 variables $t_{j,m}$ for any parameter pair in this domain; their values are not enumerated. The largest common denominator used by the principal arithmetic check has 68 decimal digits. The minimum verified normalized linear margin is

$$\frac{953977}{42042000000} > 0,$$

attained at $(n, a, k) = (34, 19, 10)$. This is a certificate upper-bound margin after normalization, not a lower bound for an unnormalized coefficient curvature.

The exact data file `certificates.json` is identified by the SHA-256 digest

a77f4fc43aec1b8eecab91c4418b6990f72dee69d802129c78efd67d2e1bcc5a.

The archived proof data and checker source are mathematical supplements to the finite theorem; the execution logs alone are not substitutes for them.

Proof of Theorem 2.1. Forests without edges were dealt with above. For a forest with an edge, choose I by Lemma 2.5 and form its exact counts $t_{j,m}$. The preceding lemmas show that these counts satisfy every structural inequality used by the certificates. Its parameters lie in the stated ranges, and the exact finite check proves (34) for every required index. The first nonpositive difference, if one occurs before $K(a)$, is followed only by nonpositive differences through index $K(a)$. The tail is nonincreasing by Lemma 2.11. Hence the whole coefficient sequence is unimodal. $\square$

2.9 The complete integer-row specification

For clarity and reproducibility, the following specifies every row family used by the finite linear certificates. A row labelled ρ means $\sum_{j,m} a_{j,m}^\rho t_{j,m} \leq b^\rho$; unspecified coefficients vanish. All indices satisfy $1 \leq j \leq v$ and $0 \leq m \leq a - j$. The symbols in the rows are defined above, and r, h, l are integers.

count(r). For $2 \leq r \leq v$, use $a_{j,m} = \mathbf{1}_{j=r}$ and $b = \binom{v}{r}$.

path(r). For $2 \leq r \leq a$, use $a_{j,m} = -\binom{m}{r-j}$ and $b = \binom{a}{r} - \binom{n-r+1}{r}$.

private(r, h). For $2 \leq r \leq v$ and $0 \leq h < a$, set

$$a_{j,m} = \mathbf{1}_{j=r}((r-1)m + a - r + 1 - rh)_+ - (v - r + 1)\mathbf{1}_{j=r-1}(m - h)_+, \quad b = 0.$$

hall(r, l). Take $0 \leq l \leq a$, $0 \leq r < \min(v, a - l)$, and $c = v - r - \max(0, l - \delta) \geq 0$. Set

$$a_{j,m} = ((r+1)\mathbf{1}_{j=r+1} - c\mathbf{1}_{j=r})\binom{m}{l}, \quad b = \begin{cases} c\binom{a}{l}, & r = 0, \\ 0, & r > 0. \end{cases}$$

The implemented domain with $l = 0$ uses $1 \leq r < v$; the omitted $(r, l) = (0, 0)$ case is the already imposed first-layer equality.

tail(r, h). For $2 \leq r \leq v$ and $0 \leq h \leq a - r + 1$, set

$$a_{j,m} = \mathbf{1}_{j=r}b_{r,h}(m) - (v - r + 1)\mathbf{1}_{j=r-1}\mathbf{1}_{m \geq h}, \quad b = 0.$$

release_upper(r, h). On the same range, set

$$a_{j,m} = s_r\mathbf{1}_{j=r-1}(m - h)_+ - \mathbf{1}_{j=r}U_{r,h}(m), \quad b = 0.$$

tail_upper(r, h). On the same range, set

$$a_{j,m} = s_r\mathbf{1}_{j=r-1}\mathbf{1}_{m \geq h} - \mathbf{1}_{j=r}V_{r,h}(m), \quad b = 0.$$

mean(r). For $2 \leq r \leq v$, set

$$a_{j,m} = -m\mathbf{1}_{j=r} - \beta_r\mathbf{1}_{j=2}, \quad b = -C_r^0 - \beta_r\binom{v}{2}.$$

union(r). For $2 \leq r \leq v$, set

$$a_{j,m} = (a - m)\left(\mathbf{1}_{j=r} - \binom{v-1}{r-1}\mathbf{1}_{j=1}\right), \quad b = 0.$$

edge_lower(r). For $2 \leq r \leq v$, set

$$a_{j,m} = -\mathbf{1}_{j=r} + \binom{v-2}{r-2} \mathbf{1}_{j=2}, \quad b = -\binom{v}{r} + \binom{v-2}{r-2} \binom{v}{2}.$$

edge_upper(r). For $2 \leq r \leq v$ and $D_0 > 0$, set

$$a_{j,m} = D_0 \mathbf{1}_{j=r} - G_r \mathbf{1}_{j=2}, \quad b = D_0 \binom{v}{r} - G_r \binom{v}{2}.$$

For $D_0 = 0$, use the **count**(r) row instead.

assumption(k). Only in a **conditional** certificate at the same index k , use

$$a_{j,m} = B_{m,j}(k), \quad b = -B_{a,0}(k).$$

These formulas specify a relaxation containing the counts of every forest after the prescribed choice of I . They do not assert that every point in the relaxation is realizable by a forest.

2.10 An explicit affine comparison at the final parameter gap

For $(n, a, k) = (60, 33, 19)$ the data include a certificate with 193 nonzero row multipliers, including one assumption row. Its denominator is $D_c = 10^7$ and its equality-weight numerator is 1395. Retaining the assumption row algebraically on the left, rather than declaring it true, gives the unconditional affine comparison

$$\Delta_{20} \leq \frac{96885639}{100000000} \Delta_{19} - \frac{4010142596414187852788421881}{28302876966000000000}. \quad (41)$$

Since the last positive fraction is strictly greater than 141686748,

$$\Delta_{20} \leq \frac{96885639}{100000000} \Delta_{19} - 141686748. \quad (42)$$

The complete weights are in `example_60_33_19.json`, which agrees exactly with the corresponding entry of the master data file. The comparison has been checked by recomputing both normalization scales and the rational residual bound. It applies simultaneously to all forests with $(n, a) = (60, 33)$, rather than to a selected graph.

2.11 Exact obstructions to extending the stated relaxations

The supplements also record two rational countermodels to specified real-valued relaxations. They are useful for delineating the reach of the method, but have a different logical status from the finite forest theorem.

The first, at $(n, a, k) = (55, 30, 18)$, satisfies all rows in the original four families **count**, **path**, **private**, and **hall** with $l \geq 1$. It has 193 nonzero count variables and a common denominator with 135 decimal digits. The second, at $(100, 55, 31)$, satisfies the extended row families above, together with 44 additional valid cover rows

$$T_r \geq \max \left(0, \binom{v}{r} - E \binom{v-2}{r-2} \right) \quad (2 \leq r \leq v). \quad (43)$$

These follow from (24) and $e \leq E$. The second countermodel has 693 nonzero count variables and a common denominator with 612 decimal digits. Its reverse-deletion rows with $s_r = 0$ need not be stored because they hold automatically for every nonnegative array.

The checker `verify_countermodels.py` regenerates the complete specified row domains, rather than trusting a list of selected rows. It checks nonnegativity, the first-layer equality, all inequalities, and the coefficient differences using integer numerators and a common positive denominator. The resulting 1 198 and 8 356 row checks, respectively, give

$$\begin{array}{c|cc} (n, a, k) & \Delta_k & \Delta_{k+1} \\ \hline (55, 30, 18) & < -69\,000\,000 & > 74\,000 \\ (100, 55, 31) & < -2 \cdot 10^{18} & > 3 \cdot 10^{17}. \end{array} \quad (44)$$

Substitution into (2) produces a formal polynomial with nonnegative rational coefficients, constant coefficient 1, linear coefficient n , and degree a . No graph realization of either array is asserted.

It follows that the explicitly specified real-valued systems cannot by themselves prove the no-recovery implication at these parameter triples: a nonnegative linear combination of valid rows must remain valid at a point satisfying all the rows. This does not disprove forest unimodality, rule out stronger consistency conditions or integrality information, or rule out a different proof at order 100. It shows precisely why replacing the upper limit 60 by 100 in the finite theorem is not justified by these certificates.

3 The monotone window and the hard-core reduction

This section fixes the interface used by the large-order chain. Its structural statements are independent of the finite-order relaxation.

Lemma 3.1 (Initial and final monotone segments). *Let F be an n -vertex forest with independence number a . Put*

$$L = \lceil n/4 \rceil, \quad U = \lfloor (n + 2a)/6 \rfloor + 1.$$

Then $c_0 \leq \dots \leq c_L$ and $c_U \geq c_{U+1} \geq \dots \geq c_a$.

Proof. Root every component. For an independent k -set S , consider triples (S, v, u) with $v \in S$ and u a child of v . If u has no child in S , send the triple to $((S \setminus \{v\}) \cup \{u\}, u)$. Otherwise choose the first child w of u belonging to S and send it to (S, w) . Outputs are independent k -sets with a marked nonroot vertex. In the first case the grandparent of the mark, if it exists, is absent; in the second it is the selected vertex v . The mark and its ancestors therefore recover the input, proving injectivity. If $D(S) = \sum_{v \in S} \deg(v)$ and $\rho_k = \sum_S |S \cap \{\text{roots}\}|$, this gives

$$\sum_S D(S) - kc_k + \rho_k \leq kc_k - \rho_k.$$

Counting extensions now yields $(k+1)c_{k+1} \geq (n-3k)c_k$. For every $k < L$, $n-3k \geq k+1$, proving the initial segment.

Fix a maximum matching of size $v = n - a$. Retaining only its edges gives a disjoint union of v pairs and $a - v$ singletons, with polynomial

$$Q(x) = (1 + 2x)^v (1 + x)^{a-v} = \sum q_k x^k.$$

Uniform independent sets in two consecutive ranks of this reference graph admit an inclusion coupling. To see this, more generally give the blocks weights $w_1, \dots, w_r > 0$. Select a k -subset of blocks proportionally to the product of its weights. If e_j denotes the elementary symmetric sum of the first $r-1$ weights, the probability of selecting the last block is $w_r e_{k-1} / (e_k + w_r e_{k-1})$. It increases with k , because the coefficient sequence of $\prod_{i < r} (1 + w_i x)$ is log-concave. Couple these last-block indicators monotonically. If they agree, couple the remaining ranks by induction; if

they differ, use the same remaining block set. Within common blocks use the same uniform element. All conditional marginals are preserved, including boundary ranks.

Independence in F is a downward-closed event under this coupling, so $c_{k+1}/q_{k+1} \leq c_k/q_k$. Finally

$$(k+1)(q_{k+1} - q_k) = (n + 2a - 6k + 1)q_k - 2(a - k + 1)(q_k - q_{k-1}).$$

For $k \geq U$ the first coefficient is nonpositive. If $q_k \geq q_{k-1}$ this proves $q_{k+1} \leq q_k$ directly; otherwise log-concavity proves it. Ratio domination transfers this to c_k . $\square$

At activity $\lambda > 0$, sample an independent set with probability $\lambda^{|S|}/I_F(\lambda)$. Let $K = |S|$, $\mu = \mathbb{E}K$, and $V = \text{Var } K$. We use the following scale-corrected mean lemma from Supplement A:

Lemma 3.2 (Activity-two mean input). *Every forest satisfies $3\mu_F(2) \geq a + 29n/60$.*

The complete proof of Lemma 3.2, including its short-leg and terminal-branch reductions, is preserved among the foundational documents of stage 900. Its last step reduces a minimal negative tree to a leaf-rooted host with branches in

$$\{T_{33}, T_{35}, P_2, P_4, P_6, P_8, P_{10}, P_{12}\}.$$

The terminal rooted trees T_{33}, T_{35} have a root and two pendant paths of lengths (3, 3) and (3, 5). Each branch T has root state $\alpha(T) - \alpha(T - \text{root}) = 0$; this is an independence-number difference, not an occupation probability. With $G(F) = 3\mu_F(2) - a - 29n/60$, host ratio $r \in [5/3, 3]$, $X = rG(H)$ and $Y = G(H - u)$, the source derives

$$E = 60(Ar + B)G(F), \quad E_2 = 180X + 120Y + 6r - 54 \geq 0,$$

where A is the product of the branch partition functions with roots deleted and $A + B$ is the product without deletion. For $2 \leq m \leq 5$ branches, all 823 multisets containing a terminal tree satisfy

$$E - \frac{A}{3}E_2 = A(60R - 100)Y + AJ(r) > 0, \quad R = (A + B)/A.$$

Here $R \geq (7/5)^2$ and the least endpoint value of J is 3523/473. For all $m \geq 6$, the remaining term is bounded below by

$$\left(\frac{7}{5}\right)^m \left(\frac{343m}{1140} - \frac{89}{60}\right) - \frac{11m}{30} + \frac{89}{60} > 0;$$

its value at 6 is 5068593/2968750 and successive increments are at least 11/60. The earlier reductions prove that these cases exhaust every minimal negative tree; the 823-case arithmetic alone does not prove that exhaustion. The dependency map identifies all three source proofs and five reconstruction scripts.

Since $d\mu/d\log \lambda = V > 0$ for a nonempty graph, the mean is strictly increasing. Conditioning on either bipartition class gives $V \geq \mu/[2(1 + \lambda)]$, whence

$$\mu(9/4) \geq \sqrt{27/26} \mu(2) > (n + 2a)/6.$$

The last inequality follows from $a \geq n/2$ and $27 \cdot 59^2 > 26 \cdot 60^2$. Since every vertex has marginal occupation at most $\lambda/(1 + \lambda)$, also $\mu(1/3) \leq n/4$. Thus every necessary rank $L \leq k < U$ has a unique activity $\lambda_k \in [1/3, 9/4]$ with $\mu(\lambda_k) = k$.

For such an activity, let π_v be the actual marginal at v , and choose an independent set B maximizing $W = \sum_{v \in B} \pi_v$. This is a maximum-weight set, not necessarily a maximum-cardinality set. Bipartiteness gives $W \geq \mu/2$. Write $C = V(F) \setminus B$. Conditional on the entire

occupied configuration in C , let Y be its size and M the number of unblocked vertices in B . Then

$$K = Y + \text{Bin}(M, q), \quad q = \frac{\lambda}{1 + \lambda}, \quad m = \mathbb{E}M, \quad qm = W. \quad (45)$$

At $\mu = k$, put $j = k - Y$ and $\delta = j - qM$. The same probability space gives

$$\mathbb{E}\delta = 0, \quad \mathbb{E}\delta^2 = V - q(1 - q)m. \quad (46)$$

No independence of M and Y is asserted.

4 The final matching-count extension: orders 59 through 79

We describe the part added by the outermost release. The older release proves the range $n \geq 80$ using the same reduction.

Lemma 4.1 (Leaves and the complement matching). *Let B be as in (45), $h = |B|$, and $H = F[C]$. Then*

$$h \geq \lceil n/3 \rceil, \quad \nu(H) \geq \max(0, n - 2h), \quad \alpha(H) \leq \lfloor n/2 \rfloor.$$

Proof. For a leaf u whose component is not a single edge, let v be its neighbor. The formulas

$$\pi_u = \frac{\lambda I_{F - \{u, v\}}(\lambda)}{I_F(\lambda)}, \quad \pi_v = \frac{\lambda I_{F - N[v]}(\lambda)}{I_F(\lambda)}$$

show $\pi_u > \pi_v$, since the second induced graph is strictly smaller than the first. Replacing v by u would improve a maximum-weight independent set; therefore $v \notin B$, and maximality implies $u \in B$. Isolated vertices are in B , and in each edge component exactly one endpoint is in B .

Let κ be the number of components of F , d the number of its edge components, $c = n - h$, $e = |E(H)|$, and s_0 the number of isolated vertices in H . Each C -vertex has a neighbor in B . An isolated vertex of H has at least two such neighbors, except for the d endpoints in edge components. Hence $e(B, C) \geq c + s_0 - d$, and $e + s_0 \leq h - \kappa + d$. The number of nontrivial components of H is $c - e - s_0 \geq n - 2h$. Choosing one edge from each gives the matching bound. Comparing twice its size with c gives $h \geq n/3$. If $h \leq n/2$, then $\alpha(H) \leq c - (n - 2h) = h$; otherwise $\alpha(H) \leq c = n - h$. $\square$

For $c \geq 2r$ define

$$D(c, r; y) = \sum_t \binom{r}{t} 2^t \binom{c - 2r}{y - t}.$$

Deleting all edges of H except a matching bounds its independent-set coefficients by those of $(1 + 2x)^r(1 + x)^{c - 2r}$. Therefore safe integer envelopes are

$$A_{n, y}^{(M)} = \max_{\substack{\lceil n/3 \rceil \leq h \leq n \\ h \geq M}} D(n - h, \max(0, n - 2h); y), \quad A_{n, y} = A_{n, y}^{(0)}. \quad (47)$$

An empty maximum is zero; binomial coefficients outside their natural ranges are zero.

Lemma 4.2 (Exact cancellation). *For every integer y ,*

$$\mathbb{E}[\mathbf{1}_{\{Y=y\}}(1 - q)^M] = \frac{i_y(H)\lambda^y}{I_F(\lambda)} \leq \frac{A_{n, y}\lambda^y}{P_n(\lambda)},$$

where $P_n(x) = \sum_t \binom{n-t+1}{t} x^t$ is the path polynomial.

Proof. The probability of an independent configuration $S \subseteq C$ is $\lambda^{|S|}(1 + \lambda)^{M(S)}/I_F(\lambda)$. Multiplication by $(1 - q)^{M(S)}$ cancels the free-vertex factor. Sum over $|S| = y$ and use (47) and the path coefficient lower bound proved in the finite-order part. $\square$

Fix a rectangle $q \in [q_a, q_b]$, $m \in [m_l, m_h]$, and $59 \leq n \leq 79$. Put $a_0 = q_a/(1 - q_a)$, $b_0 = q_b/(1 - q_b)$. The certified density bound and integer rounding give $k \geq k_{\min}$ and $m \geq k_{\min}/(2q_b)$. All real target distributions are included in

$$\mathcal{A} = \{(n, k) \in \mathbb{Z}^2 : 59 \leq n \leq 79, k_{\min} \leq k \leq \lfloor 2q_b m_h \rfloor, n/4 \leq k \leq n/2, rn \leq k \leq q_b n\}.$$

The value r here is a certified density constant, not the matching size in D . Define fixed upward rational bounds

$$B_j \geq \max_{(n,k) \in \mathcal{A}} \frac{A_{n,k-j} b_0^{k-j}}{P_n(a_0)}.$$

Terms with $k - j$ outside the range of independent-set sizes are zero. Lemma 4.2 proves $\mathbb{E}[\mathbf{1}_{\{k-Y=j\}}(1 - q)^M] \leq B_j$ throughout the rectangle. All possible conditional states belong to the finite set

$$\mathcal{S} = \{(M, j) \in \mathbb{Z}^2 : 0 \leq M \leq 79, \exists (n, k) \in \mathcal{A} : A_{n,k-j}^{(M)} > 0\}.$$

Excluding a state uses a zero combinatorial count, never a small numerical probability.

5 Rational mixture certificates and the inherited chain

Let $b_M(j; q)$ be the binomial point probability, extended by zero outside its support. Define

$$f_L = (1 - q)b_M(j; q) - qb_M(j - 1; q), \quad f_R = qb_M(j; q) - (1 - q)b_M(j + 1; q),$$

$$f_C = 2b_M(j; q) - b_M(j - 1; q) - b_M(j + 1; q).$$

Every certificate uses fixed nonnegative weights w_L, w_R, w_C , not all zero, and $f = w_L f_L + w_R f_R + w_C f_C$. Its scale S is positive. Source inequalities on the same distribution provide

$$\mathbb{E}\delta^2 \leq \alpha_0 m + \beta_0, \quad \text{Var } M \leq D_0 m, \quad \mathbb{E}(1 - q + qt_i)^M \leq e^{-\ell_i m}. \quad (48)$$

Here the finitely many $t_i \in [0, 1]$ and $\ell_i \geq 0$ are fixed rational parameters of the source bounds. Their proof is part of the chain, as described below; constants cannot be copied to a smaller threshold without checking the signs of the n -dependent terms.

The pointwise acceptance inequality is

$$Sf(M, j; q) \geq A_0 + B_0 M + C_0 \delta - D_\delta \delta^2 - D_M M^2 - \sum_i T_i (1 - q + qt_i)^M - U_j (1 - q)^M, \quad (49)$$

for every state in $\mathcal{S}$ and every q in the rectangle, with $D_\delta, D_M, T_i, U_j \geq 0$. The other coefficients may have either sign. Taking expectations gives

$$\begin{aligned} S\mathbb{E}f \geq H(m) &:= A_0 + B_0 m - D_\delta(\alpha_0 m + \beta_0) - D_M(m^2 + D_0 m) \\ &\quad - \sum_i T_i e^{-\ell_i m} - \sum_j U_j B_j. \end{aligned} \quad (50)$$

Since $H''(m) = -2D_M - \sum_i T_i \ell_i^2 e^{-\ell_i m} \leq 0$, strictly positive endpoint budgets certify all real $m \in [m_l, m_h]$.

5.1 Continuous activity certification

For a fixed state let $R(q)$ be the left side of (49) minus the right side. Write $B_L(M)$, $B_R(M)$, and $B_C(M)$ for the supplied uniform upper bounds on the second q -derivatives of the three kernels, valid for every integer j and all $q \in [q_a, q_b]$. They are reconstructed by `derivative_bounds` in stage 80's `bridge_compressed.py`, with their use specified in stage 59's proof, Section 9, and independent acceptance in `check_matching_kernel.py`. These bounds give

$$\begin{aligned} R''(q) &\leq Z_{M,j} := S(w_L B_L(M) + w_R B_R(M) + w_C B_C(M)) + 2D_\delta M^2 \\ &\quad + \sum_i T_i M(M-1)(1-t_i)^2(1-q_a + q_a t_i)^{M-2} \\ &\quad + U_j M(M-1)(1-q_a)^{M-2}. \end{aligned}$$

The terms containing $M(M-1)$ are zero when $M < 2$. With $h_q = q_b - q_a$, it suffices to check $R(q_a), R(q_b) \geq h_q^2 Z_{M,j}/8$. The interpolation remainder then proves $R(q) \geq 0$ on the entire interval. Rational enclosures for elementary functions and probabilities are directed; a precision scale is not a floating-point tolerance.

5.2 The structural source interface

For clarity, the frozen analytic sources use rooted-tree identities. If p_v is the downward occupation message, z_v its logarithmic response, and $r_v = \pi_v/p_v$, then

$$\begin{aligned} \frac{p_v}{1-p_v} &= \lambda \prod_{u \text{ child of } v} (1-p_u), \quad z_v = 1 - \sum_u p_u z_u, \\ \mu &= \sum_v p_v z_v = \sum_v p_v r_v, \quad V = \sum_v p_v r_v (1-p_v) z_v^2. \end{aligned}$$

For $\lambda \leq b$, $[1 + b(1-p_v)]^{-1} \leq r_v \leq 1$. Put

$$T_b(p) = \log \frac{b(1-p)}{p}, \quad \Lambda_b(p) = \frac{1}{p T_b(p)}, \quad h(p) = 1 - p/2.$$

If $0 \leq R_i \leq b$ and $\sum_i \log(1 + R_i) \leq t$, convexity gives

$$\sum_i R_i \leq H_b(t) := sb + e^{t-s \log(1+b)} - 1, \quad s = \lfloor t / \log(1+b) \rfloor.$$

For the displayed scalar formulas take $0 < p < q_b = b/(1+b)$. At the endpoint $p = q_b$, the vertex has no children and $z = 1$; terms containing $(1-z)^2/T_b(p)$, $(1-z)_+^2/T_b(p)$, or $(z-1)_+^2/T_b(p)$ are defined to be zero there, as are the corresponding empty-child capacity terms. The checkers handle this leaf endpoint separately. Weighted Cauchy-Schwarz over disjoint child sets gives the summed nonpositive expressions

$$\begin{aligned} \sum_v p_v [\Lambda_b(p_v)(1-z_v)_+^2 - h(p_v)(z_v)_+^2] &\leq 0, \\ \sum_v p_v [\Lambda_b(p_v)(z_v-1)_+^2 - h(p_v)(z_v)_-^2] &\leq 0, \\ \sum_v p_v \left[\frac{r_v(1-z_v)^2}{H_b(T_b(p_v))} - (1-r_v)(1-p_v)z_v^2 \right] &\leq 0. \end{aligned}$$

These, additional message-flow identities, and $\sum_v p_v(z_v - r_v) = 0$ transform the variance bounds into scalar inequalities in (p, r, z) . The response variable z ranges over all real numbers. The checkers split its three intervals $(-\infty, 0]$, $[0, 1]$, and $[1, \infty)$ and prove polynomial nonnegativity,

rather than sampling responses. Later stages replace $h(p)$ with the sharper $p/[-\log(1-p)]$ and add flow and logarithmic terms; the full formulas remain in their source documents.

For the Laplace bound in (48), hold B fixed and lower only its activities. The identity

$$\mathbb{E}e^{-s_\lambda(u)M} = \mathbb{E}e^{-uK_B}, \quad s_\lambda(u) = \log \frac{1+\lambda}{1+\lambda e^{-u}}$$

allows integration of heterogeneous nonnegative-source variance bounds. Applying a common-activity variance bound to an unverified heterogeneous path would not suffice. The source proof explicitly establishes the needed heterogeneous domain.

The basic density source at activity 1 uses the path-mean result of Andriantiana, Razanajatovo Misanantenaina, and Wagner [5]; it is not applied as a path extremality theorem at arbitrary activity. The activity-density improvements at stage 80 use 106 rational activity points. Their finite branch classes distinguish one-, two-, and three-vertex branches from larger ones; degrees 3 and 4 give 105 classes per point, while all higher degrees follow by removing one branch and an analytic induction. Thus the 11,130 checks are checks of the full induction scheme, not enumeration of forests up to some chosen order. Stage 59 adds four flow-log source certificates and recalculates the signed vertex costs with $59 \leq n \leq 79$.

5.3 Coverage and exclusion of valleys

The 231 new certificates cover 154 activity strata, subordinate to 43 parent activity intervals. Together with 188, 243, and 558 older finite kernels, respectively from stages 99, 130, and 170, they form 1220 finite rectangles. Their upper m boundary joins the stage-350 unbounded- m estimates. All 236 stage-80 kernels are excluded for $n < 80$, because some of them use signed costs requiring $n \geq 80$. This distinction is checked by the source and coverage code.

For the new certificates, the stored and replayed integer endpoint-state count is 1,185,408, and the smallest normalized expectation budget is

$$\frac{424830401847072141613691}{10^{28}} > 0.$$

The corresponding scale is 23; thus this normalized number must be divided by 23 to interpret it as a lower bound on the unscaled mixed-kernel expectation. It is not a common lower bound on a coefficient curvature.

Suppose at a necessary rank k that $c_k \leq c_{k-1}$ and $c_k \leq c_{k+1}$. Writing $p_j = c_j \lambda^j / I_F(\lambda)$, the first two inequalities give $\mathbb{E}f_L \leq 0$ and $\mathbb{E}f_R \leq 0$. They also imply

$$p_{k-1} + p_{k+1} \geq (\lambda^{-1} + \lambda)p_k \geq 2p_k,$$

so $\mathbb{E}f_C \leq 0$. This contradicts the strictly positive budget in (50). Every necessary rank therefore satisfies $c_k > \min(c_{k-1}, c_{k+1})$. If a strict descent is followed later by a strict ascent, choose the first subsequent index k with $c_{k+1} \geq c_k$. Then $c_{k-1} > c_k$ and $c_{k+1} \geq c_k$. The monotone end segments in Lemma 3.1 place such an index in $[L, U-1]$, giving a prohibited weak local minimum, including the possibility of a flat valley. Consequently all orders $59 \leq n \leq 79$ are covered once the source and coverage certificates are accepted.

The recursive replay authenticates and checks these frozen stages. The original mathematical proofs are included to justify their all-order and continuous-domain claims. Taking the base and each extension in the table establishes Proposition 1.3; this is a dependency-chain proof, not an extrapolation from the 79-vertex conditional-state bound.

6 Reproducibility, dependencies, and interpretation

The mathematical trust boundary has three parts. First, structural proofs must show that every forest satisfies the asserted inequalities. Second, arithmetic checkers must implement

Table 2: Frozen large-order stages. Each row proves the new finite interval and inherits the larger orders; the base stage includes an unbounded-order argument.

Stage	New orders	Mathematical source in Supplement A
900	$n \geq 900$	base source and kernel proof
500	$500 \leq n \leq 899$	stage 500 proof and replay
350	$350 \leq n \leq 499$	stage 350 proof and replay
170	$170 \leq n \leq 349$	stage 170 proof and replay
130	$130 \leq n \leq 169$	stage 130 proof and replay
99	$99 \leq n \leq 129$	stage 99 proof and replay
80	$80 \leq n \leq 98$	stage 80 proof and replay
59	$59 \leq n \leq 79$	final matching-count extension

those inequalities with safe rounding and verify the full parameter domains. Third, the domains must cover all required orders, ranks, activities, and conditional states. Passing one of these checks is not a substitute for the others.

The finite-order verifier constructs the full scalar domain instead of trusting a list of covered parameters. It rejects missing, duplicate, and extraneous stored certificates, inadmissible rows, and negative inequality multipliers. The large-order checkers independently reconstruct matching-count envelopes and the discrete (n, k) and (M, j) domains; continuous activity intervals and real $m = \mathbb{E}M$ intervals are certified by derivative or concavity arguments.

6.1 Commands and preserved inputs

The wrapper `verification/reproduce.py` in the distribution checks the input archive hashes, extracts into a user-selected working directory, runs the finite-order and countermodel verifiers, and, in full mode, executes the complete large-order chain. The underlying commands are:

```
python -S verify.py
python -S verify_grid.py
python -S verify_countermodels.py
python -S check_release.py
python -S replay.py
```

The first three run in Supplement B; the last two run in Supplement A. Use Python 3.10 or later and do not use `-O`. The computational proof acceptance path needs only the standard library. Optional discovery routines and numerical optimizers are not called by acceptance. The large-order chain can be much slower than the finite-order part and creates nested working directories; a short extraction path is helpful on Windows.

Fresh runs, inherited records, and diagnostic checks are separated in the accompanying audit report. A run with `-skip-prior` authenticates inherited bytes and checks the new segment, but is not described as rerunning all earlier stages. The source archive hashes and final distribution manifest identify the exact mathematical data. Runtime-dependent audit files are not frozen mathematical premises.

6.2 Precise limits of the 100-vertex obstruction

The array in `countermodel_100_55_31.json` satisfies the stated nonnegative-real relaxation for $(n, a, k) = (100, 55, 31)$, including its extra covering rows, and has $\Delta_{31} < 0 < \Delta_{32}$. This proves that those relaxation constraints alone cannot imply the needed no-valley conclusion. The formal polynomial is not asserted to be the independence polynomial of any graph.

Proposition 1.3 uses additional probabilistic and structural information, so there is no contradiction between that obstruction and the consolidated assertion. The analogous order-55 array concerns the older, weaker finite-order model.

6.3 Computational and editorial assistance

AI-assisted tools were used for mathematical review, exact computational reproduction, literature research, translation, and manuscript preparation. The resulting audit reports document their scope and limitations; they are not independent human referee reports or proof-assistant certificates. All mathematical claims are submitted for review together with the written arguments, exact certificates, and complete dependency sources. Submission instructions and provenance records are provided separately from the mathematical text.

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